FLEET TEAM DIGEST

737MAX-FTD-32-18003

Issue Title : 737MAX Multiple PSEU LRU Replacement Test Related Faults

Airplane Model	ATA	Minor Model(s)
737MAX	3209-10	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
06/21/2018	01/27/2021	01/21/2021

Estimated Completion Date	Next Update Date

Revision Description

Updated Status section. This FTD will be closed and any future updates will be addressed in 737MAX-FTD-32-18001.

Applicability

737-7/-8/-8200/-9/-BBJ

Description

As more and more 737MAX airplanes are being delivered and put into service, Boeing's diagnostic and prognostic tool called Airplane Health Management (AHM) has recorded over 200 instances of Proximity Switch Electronics Unit (PSEU) Related Flight Deck Effects (FDEs) that are perceived to be part of the "LRU Replacement Test" fault category. The AHM records the Status Messages during engine start and are displayed via the Present Leg Fault (PLF) after landing. And when on Ground, maintenance personnel typically perform the LRU Replacement Test to clear the faults.

This fault category comprises of the following Status Messages and associated FDEs:

1) Takeoff Warn Fail (FDE 31510100)

2) Takeoff Config (FDE 31510200)

3) Takeoff Configuration Horn (FDE 31510200)

4) Gear Warn Sys (FDE 32612100)

5) Gear Warn Throttle SW (FDE 32612200)

6) Altn Nose Steering (FDE 32510100)

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7) Gear Xfr Valve Control (FDE 32312100)

8) Door FWD/AFT Cargo (FDE 52713144/ 52713145)

9) Door R Eng Rly (FDE 52745142)

10)Door L/R Overwing Fltlock (FDE 52710341/52710342)

11)Door L/R Overwing (FDE 52710241/ FDE 52710242)

12)Door L/R Mid Cabin Fltlock (FDE 52710541/ 52710542)

13)Door EE Access (FDE 52714100)

This Fleet Team Digest article is written to raise fleet awareness of such fault category reported via Boeing Communication (BCS) by more than 5 operators and to provide interim actions to perform when faced with such reports of Multiple PSEU related faults pertaining to the aforementioned Status Messages.

Background

After the March 2018 737NG/MAX Fleet Team Meeting, Boeing approved a Service Related Problem (737MAX-SRP-52-0137) to authorize investigation into the Multiple PSEU LRU Replacement Test Related Fault category along with (6) other 737MAX Schedule Interruption Causing and (2) 737MAX Non-Schedule Interruption Causing fault categories.

Boeing's analysis of the group of Status messages under the LRU Replacement Test fault category indicates that the Status messages were already present prior to the flight. Initially, these Status messages were considered to be maintenance-induced (i.e., incomplete or improperly performed LRU Replacement Test). Boeing saw these same Status messages displayed during functional testing at a Boeing factory when test was aborted. Additionally, Boeing duplicated such Status messages during lab testing as part of the issue investigation. To further understand this reasoning, Boeing provides the following:

During an LRU Replacement Test, the PSEU looks for a change in the system inputs from the Landing Gear Lever, Parking Brake, Nose Wheel Steering, Alt Nose Steering, Horn Cutout, L/R Thrust Lever, EE Bay Access Door and Forward/Aft Cargo Door. When the LRU Replacement Test is aborted for whatever reason or if certain system inputs are not received by the PSEU when required, the test would fail.

To date, Boeing continues to monitor the AHM and BCS databases for further reports of PSEU Related FDEs regarding the LRU Replacement Test fault category.

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Status

Recent investigation into this fault category revealed that these LRU Replacement Test Related Faults are actually caused by the PSEU Random Access Memory (RAM) not being initialized after a power interruption. For example: Power interrupts lasting 100 - 400 milliseconds (ms) cause the PSEU to perform a “warm start” routine. During a warm start, the RAM loses power. Once power is restored to the PSEU, the RAM is not re-initialized to clear its contents, which then causes the erroneous LRU Replacement Test faults.

Boeing has completed aircraft and lab testing to validate the identified root causes and proposed solutions for this fault category. See Final Action for details of the fixes for the PSEU LRU Replacement Test Related fault category.

As of NOV-2019, System Technical Review Board (TRB) planned for late Nov-2019. Boeing will attempt to accelerate schedule, but 737-7/-8/-9 are dependent on -10 testing for certification.

As of APR-2020, all supplier and Boeing testing of the upgraded PSEU software is complete. The first PSEU -102 will be on-dock at Boeing in JUN-2020 and will be incorporated on all 737MAX aircraft at line number 7982. Boeing is developing a rework plan to install the PSEU -102 on undelivered 737MAX airplanes prior to line number 7982.

As of AUG 2020, A plan is in place to rework the undelivered Aircrafts and is being actively worked. All -7/8200 will have the -102 part as well as any undelivered -8/9 prior to LN 7918.

As of OCT 2020, For the 737-8/-9, LN7918 and on will all get the new PSEU (-102) and all the 737-8/-9 prior to LN7918, undelivered, will also get the -102.

As of JAN 2021, any future updates will be addressed in 737MAX-FTD-32-18001.

Interim Action

In order to clear the group of PSEU Related LRU Replacement Test faults, Boeing suggests the following actions:

1) Reset PSEU circuit breaker to clear faults. If faults have been cleared, no further action is required.

2) Perform the PSEU LRU Replacement Test per AMM Task 32-09-10-710-801 or the PSEU Operational Test per AMM Task

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32-09-10-740-801.

3) If faults have not been cleared, follow Fault Isolation Manual (FIM) instructions for specific maintenance message recorded.

4) Replace the PSEU, if test continues to fail.

5) Use the data loading instructions per AMM 46-13-02-970-802 to download the Airplane Data Recorder (ADR) files.

6) Provide the ADR:ALL data to Boeing via message courier to VM-FRDS with copies sent to brendan.h.o'driscoll@boeing.com and peggy.s.brown@boeing.com.

Final Action

To improve fault indication, below are the proposed solutions for this fault category:

1) Revise the PSEU software related to LRU Replacement Test: SEP-2020 (COMPLETE)

a. Reinitialize RAM following any power interruption.

b. Add a step in PSEU ground test logic to require mechanic to confirm landing gear pins are installed before test is initialized.

2) Revise the MAX Display System (MDS) software related to LRU Replacement Test: SEP-2020 (COMPLETE)

a. Modify PSEU replacement ground test “Precondition screen” to reference:

1. Landing Gear pins installation requirement

2. Warning statement indicating that failure to complete all steps in test, or stopping test, will introduce system faults.

3) Revise the Integrated Fault Isolation Manual (IFIM): DEC-2019 (COMPLETE)

a) Update troubleshooting steps for ground test related faults to include resetting PSEU circuit breaker.

S/W v01 - incorporated PSEU software update to improve LRU Replacement Test:

(1) to prevent MID EXIT faults anytime during ground tests when MID EXIT OPT is false and (2) to not set A429 output bits true until end of ground tests.

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S/W v02 - PSEU software will be updated to allow Operators to abort test without setting faults during LRU replacement test and include additional instructions to test screens.

Boeing is reviewing opportunities to improve any of the above schedule and will adjust the ECDs accordingly.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: DEC-2018 (COMPLETE)

PSEU SW v.01 Lab Testing of Initial Improvements to Door Gap Accuracy, Sensor BITE, and LRU Replacement Test: OCT-2018 (COMPLETE)

Revise SCD to update LRU Replacement Test Display Screens: FEB-2019 (COMPLETE)

PSEU SW v.02 and MDS SW Lab Testing of Final Improvements to Door Gap Accuracy, Sensor BITE, and LRU Replacement Test: MAR-2019 (COMPLETE)

MDS SW for 737-7/-8/-9: MAR-2020 (COMPLETE)

Flight Test PSEU SW v.02 and MDS SW 737-10: DEC-2019 (COMPLETE)

Completion of the Integrated Fault Isolation Manual (IFIM)/ Airplane Maintenance Manual (AMM)/ Component Maintenance Manual (CMM): MAY-2020 (TARGET)

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Production Incorporation PSEU SW v.02 and MDS SW for 737-7/-8/-9: SEPT-2020 (COMPLETE)

Supplier Service Bulletin Available PSEU SW v.02 and MDS SW for 737-7/-8/-9: SEPT-2020 (COMPLETE)

737-8200 Entry Into Service (EIS): MAR-2021 (TARGET)

Boeing Service Letter: SEPT-2021 (TARGET)

Operator Action

See Interim Action.

Part Information

The Proximity Switch Electronics Unit (PSEU) Part Number: 285A1600-101 is manufactured by British Aerospace (BAE) Systems.

Supplier Information

BAE SYSTEMS Electronics & Integrated Solutions

Commercial Electronics Product Support

Room S18 600 Main St.

Johnson City, New York 13790-1888 USA

Fax: 607-770-2318

E-mail: cs-customer.service@baesystems.com

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	AMM Task 32-09-10-710-801	Rev 4	05/15/2018
Aircraft Maintenance Manual (AMM)	AMM Task 32-09-10-740-801	Rev 4	05/15/2018
Service Related Problem (SRP)	737MAX-SRP-52-0137		
Fleet Team Digest Article	737MAX-FTD-32-18001		

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(FTD)			

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Proximity Switch Electronics Unit (PSEU)	285A1600-101	BAE Systems	

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

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